ME 374 Machine Design II
Final Exam Guidelines

The Examination is in TWO Parts.
Part A will test your breadth of knowledge in the mechanical elements covered in class.
Part B will test your ability to design a spur gear and the flexible machine elements (Belts and Chains).

Details are as follows:

Part A: You are required to answer all questions

Questions 1 and 2: Spur Gear Analysis

- Geometric Relationships
- Torque, Speed and Power Relationships
- Tooth bending stress
- Tooth contact stress

Question 3: Brake System

Actuating force
Braking torque
Condition for self-energizing

Question 4:Chain Drives

Pitch, pitch diameter
Rated horsepower calculation for standard N = 17
Corrected rated horsepower calculation for non-standard N ≠17

Question 5: Cone Clutch

Pressure, and force calculation for uniform pressure
Pressure and force calculation or uniform wear

Part B: You are required to answer Question 6 and Either Question 7 or 8

Question 6: Spur Gear Design **(REQUIRED TO ANSWER)**

Face width with factor of safety

Question 7: Chain Drive **(SELECT THIS OR Q8)**

Number teeth on both sprocket, chain number, number of strands, number of pitches, total chain length, centre-to-centre-distance, type of lubrication.

Question 8: Belt Drive **(SELECT THIS OR Q7)**

Given a belt specification, determine the number of belts needs for a power transmission.